# Haemoglobinometer



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 2 labs

**Covers these lab items:** Haemoglobin meter; Sahli's Hemoglobinometer

## What it is used for

A haemoglobinometer measures haemoglobin concentration in a blood sample using the device-specific method. Operate it according to the manufacturer's instructions and approved clinical procedures, including quality control and infection prevention.

## Main parts

- Haemoglobinometer or Sahli comparator
- Graduated tube or compatible cuvette
- Pipette, capillary, or specified blood-sampling accessory
- Reagent or diluent specified for the method
- Stirring rod or mixing accessory, if applicable
- Gloves, lancet, and approved sharps container

## Before you start

1. Verify the patient's identity and explain the test procedure.
2. Check the instrument, measuring tube, pipette, and accessories for cleanliness or damage.
3. Check reagent identity, expiry date, and storage requirements.
4. Confirm the instrument is calibrated or quality-checked as required.
5. Perform hand hygiene and prepare appropriate PPE and waste containers.
6. Collect the blood sample using the approved clinical procedure.

## How to use

1. Place the instrument on a stable, clean work surface.
2. Prepare the sample and reagent according to the specified method.
3. For a Sahli method, add the specified reagent to the graduated tube as instructed.
4. Measure the required blood volume accurately using the specified sampling accessory.
5. Add the blood to the reagent and mix according to the method instructions.
6. Allow the reaction to develop for the specified time.
7. For a visual comparator method, add diluent gradually and mix until the prescribed colour match is reached.
8. For an electronic model, insert the prepared sample or cuvette as directed by the manufacturer.
9. Read the haemoglobin result using the instrument's scale or display.
10. Record the result with the correct patient identifier, units, date, and time.
11. Report results outside the expected range or any concerning result according to clinical procedure.

2. Dispose of blood-contaminated materials and sharps in approved waste containers.
3. Clean the instrument and reusable accessories according to the manufacturer's instructions.

## ⚠ Safety

- Only trained healthcare personnel should perform or interpret the test.
- Treat all blood samples as potentially infectious and follow standard precautions.
- Use a sterile single-use lancet for each patient when capillary sampling is performed.
- Handle reagents according to their safety instructions and avoid contact with skin or eyes.
- Do not reuse disposable cuvettes, capillaries, or contaminated accessories.
- Follow quality-control procedures and report unreliable or unexpected results.

## Care & storage

- Clean and disinfect compatible surfaces after use.
- Store reagents and accessories within the specified conditions.
- Check instrument condition and perform quality control at the required frequency.
- Keep optical or comparison surfaces clean and free of scratches.
- Document maintenance, quality-control results, and equipment faults.

## Common mistakes to avoid

- Using incorrect blood volume or reagent quantity.
- Reading the result before the required reaction time or colour match.
- Using expired reagents or contaminated accessories.
- Recording results against the wrong patient or without units.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*